

02YMECHZ - Mechanics

Exam

12.2.2026 / 13:00 - 15:00

Q1: A particle of mass m can move without friction along the x -axis. At time $t = 0$, it is at rest at position $x = 0$. A force $F_x(t) = k \cos(\omega t)$ acts on it, where k and ω are constants.

- (a) Determine the acceleration $a(t)$, the velocity $v(t)$, the displacement $x(t)$. — *0.6 points*
- (b) Determine the power $P(t)$ delivered by the force as a function of time t . — *0.4 points*

Q2: Three point particles with masses $m_1 = 1$ kg, $m_2 = 2$ kg, $m_3 = 3$ kg collide instantaneously at the origin with no external force with pre-collision velocities $\vec{v}_{1i} = (2, 0, 1)$ m/s, $\vec{v}_{2i} = (-1, 1, 0)$ m/s, and $\vec{v}_{3i} = (0, -2, 3)$ m/s. Immediately *after* the collision, the particles separate, with particle 1 moves along $\hat{i}$, particle 2 along $-\hat{j}$, and particle 3 along $+\hat{k}$.

- (a) Determine the velocities after the collision: $\vec{v}_{1f}, \vec{v}_{2f}, \vec{v}_{3f}$. — *0.6 points*
- (b) Determine whether the collision is elastic or not. — *0.4 points*

Q3: The equation of motion for a driven oscillator is given as $m\ddot{x} + b\dot{x} + kx = F_0 \cos(\omega t)$, where m is the mass, b is the damping coefficient, k is the spring constant, and $F_0 \cos(\omega t)$ is the driving force. The solution for the displacement $x(t)$ of the oscillator is given by $x(t) = x_c(t) + x_p(t)$, where $x_c(t)$ is the complementary solution and $x_p(t) = A \cos(\omega t - \delta)$ is the particular solution.

- a. Describe the oscillator's behavior in the short-time ($t \approx 0$) and long-time limit ($t \rightarrow \infty$). — *0.3 point*
- b. What is the amplitude of oscillation in terms of k and F_0 when the system is driven with frequency $\omega \ll \omega_0$? What is the physical meaning of this? — *0.3 point*
- c. What is the amplitude of oscillation in terms of k , F_0 and Q when the system is driven in resonance frequency? Compared to what you found in part **b.** and considering the fact that for most of the oscillators $Q \gg 1$, what does driving the system with the resonance frequency do to the amplitude? — *0.4 points*

Given: $A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\frac{b\omega}{m})^2}}$, $Q = \frac{m\omega_0}{b}$, $\omega_0 = \sqrt{\frac{k}{m}}$

Q4: The earth rotates about its axis with angular velocity $\vec{\omega}$. The unit vectors $\hat{e}_1$, $\hat{e}_2$, and $\hat{e}_3$ are local unit vectors on the Earth's surface, where $\hat{e}_3$ is perpendicular to the surface, while $\hat{e}_1$ and $\hat{e}_2$ lie on the surface. The component of $\vec{\omega}$ along the $\hat{e}_3$ direction is ω_z . A very fast particle of mass m is projected on a frictionless horizontal surface with an initial velocity $\vec{v}_0 = v_0 \hat{e}_2$. Assume the Earth is a perfect sphere with radius R and neglect air resistance.

- a. Determine the Coriolis force by considering the general velocity of the particle $\vec{v} = v_1 \hat{e}_1 + v_2 \hat{e}_2$. Describe qualitatively the deflection due to the Coriolis force. — *0.6 points*
- b. Write the coupled equations of motion for $\hat{e}_1$ and $\hat{e}_2$ directions, and solve them for v_1 and v_2 . Does this result satisfy your initial expectations for the trajectory? — *0.4 points*

(Given: Coriolis force $\vec{F}_c = -2m\vec{\Omega} \times \vec{v}$)